

Directional Equilibrium Model (DEM Volume 2)

Dynamic Reference Trajectories, Low-Pass Memory Architectures, and Micro-Macro Cybernetic Convergence

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Abstract

This paper formalizes Volume 2 of the Directional Equilibrium Model (DEM Volume 1), expanding the framework from a static reference system to a dynamic, time-variant cybernetic control model. Volume 1 established the micro-scale cycle— $3 \rightarrow 7^{\wedge} \rightarrow 6 \rightarrow 9 \rightarrow 3$ —operating within a fixed reference frame N . Volume 2 advances this architecture by formalizing scale separation: micro-scale instantaneous convergence versus macro-scale temporal evolution.

Within this paradigm, the reference parameter N is re-engineered as a continuous macro-variable $\mathbf{N}(t)$, dynamically shaped by systemic memory $M(t)$. Memory functions as a first-order low-pass filter operating on the discrete sequence of stabilized micro-states $\mathbf{x}_9(k)$. High-frequency noise and transient perturbations inherent to micro-level adjustments (6) are attenuated, preserving structural directional inertia while allowing the reference frame to track non-stationary environmental manifolds. Furthermore, this paper formalizes decision dynamics under hard temporal execution constraints (T_{act}), modeling trajectory selection as an optimal control problem that balances micro-convergence fidelity against time-decay penalties.

Diagnostic failure modes are mathematically defined, specifically phase lag, reference aliasing, and non-convergent resonance. Volume 2 transforms the Directional Equilibrium Model into a rigorous, closed-loop state-space architecture capable of dynamic orientation, adaptive memory-driven reference update, and optimal action selection in non-stationary environments.

1 Introduction

Modern control theory, decision architecture, and signal processing models frequently struggle with the dual requirement of short-term disturbance rejection and long-term adaptive tracking. Static optimization frameworks establish fixed objective functions, which become invalid under non-stationary environmental dynamics. Conversely, hyper-adaptive models that continuously update objectives based on instantaneous inputs suffer from high-frequency instability, noise tracking, and loss of directional coherence.

Volume 1 of the Directional Equilibrium Model established the fundamental mechanics of reference-driven orientation. It defined N as the center bearing—the structural condition mak-

ing meaningful signal evaluation possible—and introduced the micro-scale iterative loop:

$$3 \longrightarrow 7^\wedge \longrightarrow 6 \longrightarrow 9 \longrightarrow 3 \quad (1)$$

While Volume 1 assumed a stationary reference frame to isolate the mechanics of convergence, real-world cybernetic systems operate across multi-scale temporal horizons. A fixed N in a shifting environment inevitably introduces systemic bias, leading to model mismatch, uncorrected state error, and inevitable non-convergence.

Volume 2 resolves this limitation by formalizing the dynamic evolution of $\mathbf{N}(t)$ through time-domain scale separation. The core architecture splits processing into two distinct operational layers:

- **Micro-Scale Convergence:** Rapid, instantaneous signal extraction, directional adjustment, and stabilization ($3 \rightarrow 7^\wedge \rightarrow 6 \rightarrow 9$) occurring over a characteristic timeframe τ_μ .
- **Macro-Scale Evolution:** The continuous adaptation of the reference frame $\mathbf{N}(t)$ over a macro-timeframe $\tau_M \gg \tau_\mu$, mediated by a low-pass memory transfer function operating on prior stabilized micro-states $\mathbf{x}_9(k)$.

By modeling memory as an attenuation filter, the system filters out stochastic micro-variations while integrating persistent structural shifts. This creates a dynamically stable reference trajectory against which actions can be evaluated and executed under explicit temporal constraints (T_{act}).

2 Core Model Definition and Scale Separation

The Volume 2 architecture operates via a multi-scale feedback topology. The static reference frame N from Volume 1 is replaced by the time-dependent reference state $\mathbf{N}(t) \in \mathbb{R}^d$.

2.1 Micro-Scale State Variables (τ_μ)

The micro-scale loop operates as a discrete-continuous hybrid system, mapping distributed input spaces to stabilized local attractors:

- $\mathbf{x}_3(k) \in \mathbb{R}^d$ — **Input Space (Distributed Data Field):** A vector-valued stochastic distribution representing unfiltered observations, raw state measurements, or candidate trajectories at iteration k .
- $\mathbf{s}_{7^\wedge}(k) \in \mathbb{R}^d$ — **Emergent Signal Synthesis:** A non-linear manifold projection operator $\mathcal{F}_{7^\wedge}$ that collapses distributed inputs into a coherent candidate state vector:

$$\mathbf{s}_{7^\wedge}(k) = \mathcal{F}_{7^\wedge}(\mathbf{x}_3(k)) \quad (2)$$

- $\mathbf{x}_6(t) \in \mathbb{R}^d$ — **Directional State Vector:** The continuous-time trajectory during directional evaluation and adjustment relative to the instantaneous reference frame $\mathbf{N}(t)$.
- $\mathbf{x}_9(k) \in \mathbb{R}^d$ — **Micro-Stabilized State:** The fixed-point attractor achieved at the conclusion of micro-adjustment, defined where state velocity drops below an equilibrium threshold ϵ_9 :

$$\mathbf{x}_9(k) = \mathbf{x}_6(t_{\text{stab}}) \quad \text{where} \quad \left\| \frac{d\mathbf{x}_6}{dt} \Big|_{t=t_{\text{stab}}} \right\| \leq \epsilon_9 \quad (3)$$

2.2 Macro-Scale State Variables (τ_M)

- $M(t) \in \mathbb{R}^d$ — **Integrated Memory State:** The accumulated, time-weighted history of prior micro-stabilizations $\mathbf{x}_9(k)$.
- $\mathbf{N}(t) \in \mathbb{R}^d$ — **Dynamic Reference Trajectory:** The moving center bearing of the system, continuously updated by the memory low-pass filter to define the current origin for micro-scale directional adjustment.

3 Micro-Scale Convergence Mechanics ($3 \rightarrow 7^\wedge \rightarrow 6 \rightarrow 9$)

Micro-scale convergence governs instantaneous state optimization within a given iteration k , operating under a fixed reference snapshot $\mathbf{N}_k = \mathbf{N}(t_k)$.

3.1 Manifold Collapse ($7^\wedge$)

The projection operator $\mathcal{F}_{7^\wedge}$ performs non-linear dimensionality reduction and noise suppression across the input distribution $\mathbf{x}_3(k)$. Formally, if $\mathbf{x}_3(k)$ is defined by M discrete samples $\{\xi_1, \xi_2, \dots, \xi_M\}$, the emergent signal $\mathbf{s}_{7^\wedge}(k)$ is the density peak extracted via kernel manifold projection:

$$\mathbf{s}_{7^\wedge}(k) = \arg \max_{\mathbf{z}} \sum_{i=1}^M K_h(\mathbf{z} - \xi_i) \quad (4)$$

where K_h is a smoothing kernel of bandwidth h . This represents the structural collapse of input variance into a singular candidate vector.

3.2 Directional Mechanics (6) in Continuous State Space

The adjustment phase 6 governs the time-evolution of the state vector $\mathbf{x}_6(t)$ driven by a potential function $V(\mathbf{x}_6, \mathbf{N}_k)$ anchored at the reference frame $\mathbf{N}_k$:

$$V(\mathbf{x}_6, \mathbf{N}_k) = \frac{1}{2} (\mathbf{x}_6 - \mathbf{N}_k)^T \mathbf{Q} (\mathbf{x}_6 - \mathbf{N}_k) \quad (5)$$

where $\mathbf{Q} \succ 0$ is a positive-definite symmetric weighting matrix establishing metric space scales. The continuous differential state update is governed by:

$$\frac{d\mathbf{x}_6(t)}{dt} = -\nabla_{\mathbf{x}_6} V(\mathbf{x}_6(t), \mathbf{N}_k) + \mathbf{u}_{\text{dir}}(t) = -\mathbf{Q} (\mathbf{x}_6(t) - \mathbf{N}_k) + \mathbf{u}_{\text{dir}}(t) \quad (6)$$

The control term $\mathbf{u}_{\text{dir}}(t)$ regulates the bidirectional adjustment mechanism:

1. **Inward Dynamics (Contraction/Refinement):** Operates when $\mathbf{u}_{\text{dir}}(t) = -\mathbf{K}_p \mathbf{e}(t)$, where $\mathbf{e}(t) = \mathbf{x}_6(t) - \mathbf{N}_k$. The dynamics are purely dissipative, driving the system strictly toward $\mathbf{N}_k$:

$$\frac{d\mathbf{x}_6(t)}{dt} = -(\mathbf{Q} + \mathbf{K}_p) (\mathbf{x}_6(t) - \mathbf{N}_k) \quad (7)$$

The derivative of the Lyapunov function $\dot{V}(\mathbf{x}_6) < 0$ guarantees asymptotic stability toward $\mathbf{N}_k$.

2. **Outward Dynamics (Expansion/Exploration):** Operates when local convergence stalls or input diversity is inadequate. The control term injects structured exploratory velocity $\mathbf{\Gamma}_{\text{exp}}(t)$:

$$\mathbf{u}_{\text{dir}}(t) = \mathbf{\Gamma}_{\text{exp}}(t) \quad \text{such that} \quad \langle \mathbf{\Gamma}_{\text{exp}}(t), \mathbf{e}(t) \rangle > 0 \quad (8)$$

This forces the state vector to expand outward from $\mathbf{N}_k$, exploring adjacent state space manifolds to collect higher-variance data for the subsequent input field (3).

4 Macro-Evolution and Low-Pass Memory Architecture

The transition from a static reference N to a dynamic trajectory $\mathbf{N}(t)$ requires a continuous mapping from discrete micro-stabilizations $\mathbf{x}_9(k)$ to the macro reference state.

4.1 Continuous Memory Convolution

Memory $M(t)$ acts as a causal impulse response filter $h_M(t)$ applied to the continuous pulse-train representation of stabilized micro-states $\mathbf{x}_9(t) = \sum_k \mathbf{x}_9(k) \delta(t - t_k)$:

$$\mathbf{N}(t) = (h_M * \mathbf{x}_9)(t) = \int_{-\infty}^t h_M(t - \tau) \mathbf{x}_9(\tau) d\tau \quad (9)$$

To enforce a low-pass filtering characteristic, $h_M(t)$ is defined as a first-order exponential decay kernel with memory time-constant τ_m :

$$h_M(t) = \frac{1}{\tau_m} e^{-\frac{t}{\tau_m}} U(t) \quad (10)$$

where $U(t)$ is the Heaviside step function.

4.2 State-Space and Differential Formulation

In the time domain, the dynamic evolution of the reference frame $\mathbf{N}(t)$ is governed by the linear differential equation:

$$\tau_m \frac{d\mathbf{N}(t)}{dt} + \mathbf{N}(t) = \mathbf{x}_9(t) \quad (11)$$

Transforming to the Laplace domain yields the transfer function $H_M(s)$:

$$H_M(s) = \frac{\mathbf{N}(s)}{\mathbf{X}_9(s)} = \frac{1}{\tau_m s + 1} = \frac{\omega_c}{s + \omega_c} \quad (12)$$

where $\omega_c = \frac{1}{\tau_m}$ defines the system's spatial cutoff frequency.

4.3 Discrete Recurrence Formulation

For computational implementation across discrete micro-cycles k occurring at sampling intervals $\Delta t = t_{k+1} - t_k$, the continuous update is discretized via Bilinear or Exponential Euler transformation:

$$\mathbf{N}(k+1) = (1 - \alpha) \mathbf{N}(k) + \alpha \mathbf{x}_9(k) \quad (13)$$

where the smoothing coefficient $\alpha \in (0, 1]$ is strictly defined by the ratio of sampling period to memory time-constant:

$$\alpha = 1 - e^{-\frac{\Delta t}{\tau_m}} \quad (14)$$

4.4 Filtering Properties of Memory

- **High-Frequency Suppression** ($\omega > \omega_c$): Micro-scale perturbations, sensor noise, and transient exploratory state adjustments occurring at frequencies higher than ω_c are attenuated at -20 dB/decade. This prevents $\mathbf{N}(t)$ from tracking transient noise.
- **Low-Frequency Pass-Through** ($\omega < \omega_c$): Persistent, structural changes in the location of stabilized micro-states $\mathbf{x}_9(k)$ pass through the filter, shifting the center bearing $\mathbf{N}(t)$ smoothly along the true environmental trajectory.

5 Trajectory Selection (Decision) Under Temporal Constraints

A system cannot iterate indefinitely in real-time environments. Action execution is bounded by an explicit time deadline T_{act} . Trajectory selection is formulated as an optimal control problem over a finite horizon $t \in [0, T_{\text{act}}]$.

5.1 Optimal Control Formulation

Let the chosen execution trajectory be $\mathbf{x}(t)$ for $t \in [0, T_{\text{act}}]$, controlled by adjustment force $\mathbf{u}_6(t)$. The goal is to minimize a cost functional J balancing directional error relative to $\mathbf{N}(t)$, control effort $\mathbf{u}_6(t)$, and terminal alignment at T_{act} :

$$\min_{\mathbf{u}_6(t)} J = \int_0^{T_{\text{act}}} \left(\|\mathbf{x}(t) - \mathbf{N}(t)\|_{\mathbf{Q}}^2 + \|\mathbf{u}_6(t)\|_{\mathbf{R}}^2 \right) dt + \|\mathbf{x}(T_{\text{act}}) - \mathbf{N}(T_{\text{act}})\|_{\mathbf{P}}^2 \quad (15)$$

subject to the physical state dynamics:

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{f}(\mathbf{x}(t), \mathbf{u}_6(t)), \quad \mathbf{x}(0) = \mathbf{s}_{7^\wedge}(k) \quad (16)$$

where $\mathbf{Q} \succeq 0$, $\mathbf{R} \succ 0$, and $\mathbf{P} \succeq 0$ are weighting matrices for path tracking error, control energy, and terminal state error respectively.

5.2 Temporal Trade-Off Dynamics

The decision to execute an action at time T_{act} introduces a fundamental trade-off between convergence accuracy and temporal execution delay:

1. **Iteration-Dominated Regime** ($T_{\text{act}} \gg \tau_\mu$): The system has sufficient temporal budget to execute K complete micro-cycles ($3 \rightarrow 7^\wedge \rightarrow 6 \rightarrow 9$), achieving local convergence $\mathbf{x}(T_{\text{act}}) \approx \mathbf{x}_9(K)$ near the global minimum of $V(\mathbf{x}, \mathbf{N})$.
2. **Time-Constrained Regime** ($T_{\text{act}} \sim \tau_\mu$): The deadline forces execution during the active adjustment phase 6 before reaching stabilization 9. The executed action vector is:

$$\mathbf{x}_{\text{exec}} = \mathbf{x}_6(T_{\text{act}}) \quad (17)$$

The system accepts a non-zero residual alignment error $\mathbf{e}(T_{\text{act}}) = \mathbf{x}_6(T_{\text{act}}) - \mathbf{N}(T_{\text{act}})$ to satisfy the temporal constraint.

6 System Stability, Phase Lag, and Diagnostic Failure Modes

The dynamic coupling between the micro-loop ($3 \rightarrow 7^\wedge \rightarrow 6 \rightarrow 9$) and macro-evolution ($\mathbf{N}(t)$) introduces specific failure modes derived from control-loop dynamics.

Table 1: System Diagnostic Matrix for Volume 2 Cybernetic Control

Diagnostic Failure Mode	Mathematical Condition	Physical System Symptom
Reference Phase Lag	$\tau_m \gg \frac{1}{\omega_E}$	Systemic hysteresis, delayed adaptation.
Reference Aliasing	$\tau_m \ll \Delta t$	Instability, noise tracking, loss of center bearing.
Non-Convergent Resonance	$\Re(\lambda_i) \geq 1$	Persistent oscillation, unbounded cycle limit.

6.1 Reference Phase Lag (Hysteresis)

- **Condition:** The memory time-constant τ_m is set excessively large relative to the rate of environmental change ω_E ($\tau_m \gg \frac{1}{\omega_E}$).
- **Dynamics:** The transfer function $H_M(s)$ introduces a phase shift $\phi(\omega) = -\arctan(\omega\tau_m)$. At environmental frequency ω_E , the reference frame lags behind the true equilibrium manifold:

$$\Delta \mathbf{N}(t) = \mathbf{N}(t) - \mathbf{N}_{\text{true}}(t) \approx -\tau_m \frac{d\mathbf{N}_{\text{true}}(t)}{dt} \quad (18)$$

- **Symptom:** The system exhibits systemic hysteresis. Micro-adjustments (6) converge toward an obsolete reference point, generating persistent directional bias and constant tracking error.

6.2 Reference Aliasing (Over-Adaptation / Instability)

- **Condition:** The memory time-constant is set too small ($\tau_m \rightarrow 0$, or $\alpha \rightarrow 1$).
- **Dynamics:** The cutoff frequency $\omega_c \rightarrow \infty$, removing the filtering property of $H_M(s)$. High-frequency noise and transient outward exploratory states $\mathbf{x}_6(t)$ are directly absorbed into $\mathbf{N}(t)$:

$$\mathbf{N}(k+1) \approx \mathbf{x}_9(k) \quad (19)$$

- **Symptom:** The reference frame loses its function as a center bearing. $\mathbf{N}(t)$ oscillates wildly, tracking instantaneous input noise rather than underlying structural signals. System stability breaks down.

6.3 Non-Convergent Micro-Macro Resonance

- **Condition:** Feedback coupling between micro-stabilization $\mathbf{x}_9(k)$ and macro-reference update $\mathbf{N}(k)$ creates positive feedback at a resonant frequency ω_R .
- **Dynamics:** Consider the linear system coupled state update:

$$\begin{bmatrix} \mathbf{x}_9(k+1) \\ \mathbf{N}(k+1) \end{bmatrix} = \begin{bmatrix} \mathbf{A}_6 & \mathbf{B}_6 \\ \alpha \mathbf{A}_6 & (1-\alpha)\mathbf{I} + \alpha \mathbf{B}_6 \end{bmatrix} \begin{bmatrix} \mathbf{x}_9(k) \\ \mathbf{N}(k) \end{bmatrix} \quad (20)$$

If the real part of any eigenvalue of the system matrix satisfies $\Re(\lambda_i) \geq 1$, the closed-loop system is unstable.

- **Symptom:** Instead of converging to equilibrium, the micro-state $\mathbf{x}_9$ and reference frame $\mathbf{N}$ execute expanding limit cycles or unbounded oscillations, causing total loss of system orientation.

7 Recollection, Branching, and Re-Centering Dynamics

When diagnostic monitoring reveals non-convergence or reference tracking breakdown, the system transitions from iterative continuation to branching and re-centering.

7.1 Divergence Detection Threshold

At each stabilization step 9, the system evaluates the innovation error $\mathbf{e}_k = \mathbf{x}_9(k) - \mathbf{N}(k)$. Branching is triggered when the Mahalanobis distance exceeds a critical diagnostic threshold δ_{crit} :

$$D_M(\mathbf{x}_9(k), \mathbf{N}(k)) = \sqrt{(\mathbf{x}_9(k) - \mathbf{N}(k))^T \Sigma_N^{-1} (\mathbf{x}_9(k) - \mathbf{N}(k))} > \delta_{\text{crit}} \quad (21)$$

where Σ_N is the covariance matrix of historical reference variations.

7.2 Re-Centering Mechanics

Upon exceeding δ_{crit} , the system executes a structural reset:

1. **Memory Kernel Flushing:** The memory state $M(t)$ is cleared, or the cutoff frequency is temporarily widened ($\tau_m \rightarrow \tau_{\text{fast}} \ll \tau_m$) to allow rapid reference re-alignment.
2. **Input Space Re-Collection ($\mathbf{3}_{\text{new}}$):** The prior stabilized output $\mathbf{x}_9(k)$ is isolated from the new collection layer $\mathbf{3}_{\text{new}}$. The input field is resampled over a broader spatial boundary to construct an uncontaminated input manifold.
3. **Reference Re-Initialization:** $\mathbf{N}_{\text{new}}$ is assigned directly to the principal component mode of $\mathbf{3}_{\text{new}}$, establishing a re-centered bearing for subsequent micro-scale operations:

$$\mathbf{N}_{\text{new}} = \arg \max_{\mathbf{v}} \text{Var}(\langle \mathbf{v}, \mathbf{x}_{\mathbf{3}_{\text{new}}} \rangle) \quad (22)$$

8 Applications and Cybernetic Systems Implementation

The formalisms of Volume 2 provide actionable implementations across complex computational domains.

8.1 Autonomous Control and Robotics under Dynamic Disturbances

In physical autonomous systems (e.g., aerial robotics navigating turbulent velocity fields):

- $\mathbf{x}_3(k)$ represents multi-modal sensor streams (LiDAR, IMU, optical flow).
- $\hat{\mathbf{x}}$ computes instantaneous state estimates via nonlinear Kalman filtering.
- $\mathbf{u}$ calculates control surfaces effort via Model Predictive Control (MPC).
- $\mathbf{N}(t)$ acts as the dynamic path center bearing, low-pass filtering high-frequency wind gust disturbances via $H_M(s)$ to preserve the smooth global flight plan.

8.2 Multi-Agent Adaptive Systems and Distributed Consensus

In distributed network consensus:

- Agents perform local micro-convergence cycles ($3 \rightarrow 7^\wedge \rightarrow 6 \rightarrow 9$) to process local telemetry.
- Macro-memory filtering across the network computes the temporal reference trajectory $\mathbf{N}(t)$, ensuring global network orientation without requiring high-bandwidth synchronization of every local micro-perturbation.

8.3 Signal Processing and Adaptive Filtering

In non-stationary signal processing:

- The architecture functions as an adaptive notch/pass filter where $\mathbf{N}(t)$ tracks the time-varying mean of non-stationary signals, while the micro-loop isolates and processes high-frequency transient anomalies relative to that moving baseline.

9 Discussion and Operational Limitations

Volume 2 expands the Directional Equilibrium Model into a fully dynamic mathematical framework. However, several system parameter constraints and operational limitations must be noted:

1. **Parameter Tuning Sensitivity:** The performance of the system relies heavily on the proper calibration of three core parameters: the memory time-constant τ_m , the divergence threshold δ_{crit} , and the decision deadline T_{act} . Improper scale separation ($\tau_m \approx \tau_\mu$) degrades the system to either reference lag or noise tracking.
2. **Computational Complexity of Manifold Collapse ($7^\wedge$):** Non-linear kernel density estimations and high-dimensional projections at phase $7^\wedge$ incur significant $\mathcal{O}(M^2d)$ computational overhead per micro-cycle, which can restrict performance in ultra-low-latency processing environments.
3. **Non-Gaussian Noise Fields:** The linear low-pass formulation of memory $H_M(s) = \frac{1}{\tau_m s + 1}$ assumes bounded, zero-mean noise characteristics. In environments dominated by heavy-tailed, non-Gaussian noise distributions, linear filtering is sub-optimal, requiring non-linear or rank-order memory filter kernels.

10 Conclusion

Volume 2 of the Directional Equilibrium Model establishes a formal mathematical and cybernetic framework for multi-scale system navigation. By transitioning N from a static center bearing to a dynamic trajectory $\mathbf{N}(t)$ modulated by low-pass memory mechanics ($H_M(s)$), the model achieves functional scale separation: micro-scale noise suppression and local state stabilization are decoupled from macro-scale reference evolution.

Furthermore, the integration of optimal trajectory selection under temporal constraints (T_{act}) provides a rigorous control-theoretic foundation for decision-making under execution deadlines. Diagnostic criteria derived from phase lag, aliasing, and spectral analysis enable self-assessment

and automated re-centering. Ultimately, Volume 2 transforms the Directional Equilibrium Model into a robust, closed-loop state architecture capable of maintaining structural orientation, filtering environmental noise, and executing optimal actions within dynamic, complex systems.